using pb_tool for qgis plugin building, what is the workflow?

i have an empty plugin after using "create".
do i have to "compile" first and then "deploy" or can i just straight "deploy" and then refresh with plugin reloader?

With **pb_tool** for building **QGIS** plugins, the workflow depends on whether your plugin uses **compiled UI/resources** or only pure Python files.

Short answer

- If you **don't have** `.ui` **or** `.qrc` **changes**, you can **deploy directly**.
- If you **changed** `.ui` **or** `.qrc`, you should **compile first**, then **deploy**.

Typical development loop:

1. Edit Python / UI files
2. `pb_tool compile` *(only if UI/resources changed)*
3. `pb_tool deploy`
4. Reload plugin using **Plugin Reloader**

Typical pb_tool workflow

1. Create plugin

Bash

This scaffolds the plugin and generates **pb_tool.cfg**.

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2. Develop your plugin

Edit files such as:

```
plugin.py
plugin_dialog.py
```

```
plugin_dialog_base.ui
resources.qrc
```

---

3. Compile (only when needed)

Run:

```bash
```

This converts:

Source	Output
`.ui`	`.py` (via `pyuic`)
`.qrc`	`_rc.py` (via `pyrcc`)

Example:

```
plugin_dialog_base.ui → plugin_dialog_base.py
resources.qrc → resources_rc.py
```

If you **didn't modify UI or resources**, you can skip this.

---

4. Deploy to QGIS plugin folder

```bash
```

This copies the plugin into the **QGIS plugin directory**, typically:

```
~/.local/share/QGIS/QGIS3/profiles/default/python/plugins/
```

(or equivalent on your OS)

---

5. Reload inside QGIS

Use **Plugin Reloader**:

1. Open QGIS
2. Click **Reload Plugin**
3. Your changes are loaded instantly

No restart needed.

Typical development loop (recommended)

Most developers do:

```bash
```

after editing Python files, then reload the plugin.

Only run:

```bash
```

when you modify:

- `.ui`
- `.qrc`

Faster workflow (many devs use this)

Common loop:

```
edit code
pb_tool deploy
reload plugin
```

Compile only occasionally.

💡 **Pro tip:**

You can combine commands:

Bash

or make an alias:

Bash

If you want, I can also show the **best folder structure for QGIS plugins using pb_tool** (many templates generate messy layouts).

pyuic5 is not in your path---unable to compile your ui files
pyrcc5 is not in your path---unable to compile your resource file(s)

That error means `pb_tool` **cannot find the Qt build tools** used to convert UI/resource files:

- `pyuic5` → converts `.ui` → `.py`
- `pyrcc5` → converts `.qrc` → `_rc.py`

These tools come with **PyQt5**, which is bundled inside **QGIS**, but often **not on your system PATH**.

Quick fixes

Option 1 (Recommended): Use the QGIS Python environment

Run commands inside the QGIS environment where `pyuic5` already exists.

Linux (often works automatically)

Check:

Bash

If not found, try:

Bash

Option 2: Install PyQt5 tools in your Python environment

If you're running `pb_tool` with your **system Python**, install:

Bash

Then verify:

Bash

Option 3: Point pb_tool to the correct tools (common with QGIS installs)

Edit `pb_tool.cfg`:

```
[compile]
```

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Typical locations:

Linux (APT QGIS install)

```bash
/usr/bin/pyrcc5
```

macOS (QGIS bundle)

```
/Applications/QGIS.app/Contents/MacOS/bin/pyuic5
```

Windows (OSGeo4W)

```
C:\OSGeo4W\apps\Python39\Scripts\pyuic5.exe
```

Quick test

Run:

```bash
```

If they respond, `pb_tool compile` will work.

Important: You can still develop without compiling

If your plugin:

- has **no** `.ui` **files**
- has **no** `.qrc` **files**

then you can **skip compile entirely** and just:

```bash
```

and reload in QGIS.

💡 **Tip:** Many QGIS plugin developers avoid `.ui` compilation entirely by loading UI dynamically:

Python

Then `pyuic5` **is never required** during development.

If you'd like, I can also show the **modern QGIS plugin workflow that avoids** `pb_tool compile` **completely** (much smoother for development).

troll4hire@lapos-ideapad-320:~/Documents/_python_/PotreeCraft/qgis_plugin$ pb_tool deploy
Cannot mix incompatible Qt library (5.15.3) with this library (5.15.2)
Aborted (core dumped)

Thought for 25s

That crash is **not a deploy-order problem**. It means `pb_tool` is being run with a **different Qt/PyQt stack** than the one your QGIS install is using, and `pb_tool deploy` tries to import Qt-related pieces during its build/deploy path. `pb_tool deploy` normally **does compile as part of deploy** unless you use `-q/--quick`, so the Qt mismatch can surface even on deploy. G Sherman +1

In your case, the important clue is:

- one side is **Qt 5.15.3**
- the other side is **Qt 5.15.2**

That exact class of error happens when libraries from two different Qt builds get mixed in one process. QGIS has a documented example of the same failure when importing QGIS from a “standard Python console” on Ubuntu, which is very similar to running `pb_tool` from the wrong Python environment. GitHub

What to do

The fix is to make sure `pb_tool`, `PyQt5`, **and QGIS all come from the same environment/source**.

Best approach on Linux

Use the **system/QGIS Python environment**, not a pip environment with its own Qt wheels.

That usually means:

1. **Do not run** a `pb_tool` installed into a venv/conda env that also has pip-installed `PyQt5`.
2. Install `pb_tool` into the same Python that QGIS uses, or use distro packages for the Qt tools.
3. Avoid mixing:
   - apt/QGIS Qt libraries
   - pip `PyQt5` / `PyQt5-Qt5`

That mix is the usual cause of this error. The pb_tool docs say it is a Python CLI installed with `pip`, and deploy includes compile unless you use `--quick`; the QGIS issue shows that using the "standard Python" side can trigger the mismatch. G Sherman +1

Practical recovery steps

First, check which `pb_tool` you are actually running:

```bash
```

Then check whether pip-installed PyQt is present in that same environment:

```bash
```

If you see pip-installed `PyQt5` or `PyQt5-Qt5`, that is the first thing I would remove:

```bash
```

Then install the distro/QGIS-side tools instead:

Bash

And install `pb_tool` against the system Python, not a separate Qt-bearing env:

Bash

That keeps `pb_tool` lightweight while letting Qt come from the same apt/QGIS side. The pb_tool docs confirm pip install for the tool itself, and Linux reports around this error consistently point to mixed Qt libraries in the runtime path. G Sherman +2

Easiest workaround right now

Since your plugin is empty, try:

Bash

`--quick` skips compiling UI/resources/docs/translations during deploy. That option is explicitly documented by pb_tool. If your plugin currently has only plain Python files, this may let deployment work immediately. G Sherman

If `--quick` works, your problem is definitely in the Qt-backed compile/build part, not the basic file copy.

Recommended dev loop after that

For day-to-day work:

- Python-only edits:

  Bash

  then reload in Plugin Reloader.
- After changing `.ui` or `.qrc`:
  fix the environment issue first, then use

  Bash

```
pb_tool deploy
```

Likely root cause in one sentence

You are probably launching `pb_tool` from a Python install that has **pip Qt 5.15.2** while QGIS/Ubuntu is providing **Qt 5.15.3**, or the reverse, and Qt refuses to load both in one process. GitHub +1

Paste the output of these three commands and I'll tell you exactly which side is mismatched:

```bash
which pb_tool
```

Sources

troll4hire@lapos-ideapad-320:~/Documents/_python_/PotreeCraft/qgis_plugin$ pb_tool deploy
Your plugin path does not exist. Make sure your profile directory is present: (/home/troll4hire/.local/share/QGIS/QGIS3/profiles/default/python/plugins)
Profiles are created from the Settings->User Profiles menu.
Unable to determine where to deploy your plugin

Thought for 13s

`pb_tool` is now past the Qt issue and failing for a simpler reason: it cannot find your **active QGIS profile's plugin folder**. QGIS stores external Python plugins under the `python/plugins` directory of the **active user profile**. QGIS +1

What this means in practice:

- `pb_tool` is trying to deploy to `/home/troll4hire/.local/share/QGIS/QGIS3/profiles/default/python/plugins`
- that path does not exist yet
- QGIS profiles are created from **Settings → User Profiles**, and the active profile folder can be opened from there. QGIS +1

Fastest fix

Open QGIS once, then go to:

Settings → User Profiles → Open Active Profile Folder

That will show the real profile directory QGIS is using. Inside it, make sure this exists:

Bash

If it does not, create it:

Bash

Then run:

Bash

`pb_tool` supports deploy to the QGIS plugins directory, and `--quick` skips compile/build steps during deploy. G Sherman +1

Better check

Your active profile may **not** be `default`. QGIS uses the active profile's path, not necessarily `default`. The docs say external plugins live under the active profile path. QGIS

So verify the profile name in QGIS first. If your active profile is something like `testing`, the correct path would be more like:

Bash

Most robust fix with `pb_tool.cfg`

You can also bypass auto-detection and set the deploy path explicitly in `pb_tool.cfg`. The config supports a full path for where the plugin directory should be copied. GitLab

Set it to the exact plugin folder for your active profile, for example:

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Then deploy again.

Recommended workflow from here

1. Launch QGIS and confirm the active profile folder. QGIS +1
2. Create `python/plugins` if missing.
3. Set `plugin_path` in `pb_tool.cfg` if auto-detection keeps guessing wrong. GitLab
4. Use:

Bash

for Python-only edits, then reload with Plugin Reloader. `pb_tool` is intended to deploy to your plugins directory for testing. G Sherman

If you paste your `pb_tool.cfg`, I'll tell you exactly what line to change.

Sources

```
troll4hire@lapos-ideapad-320:~/Documents/_python_/PotreeCraft/qgis_plugin$ pb_tool deploy
Deploying to /home/troll4hire/.local/share/QGIS/QGIS3/profiles/default/python/plugins/potreecraft 2.0
Deploying will:
        * Remove your currently deployed version
        * Compile the ui and resource files
```

```
        * Build the help docs
        * Copy everything to your
/home/troll4hire/.local/share/QGIS/QGIS3/profiles/defau
lt/python/plugins/potreecraft 2.0 directory

Proceed? [y/N]: y
Removing plugin from
/home/troll4hire/.local/share/QGIS/QGIS3/profiles/defau
lt/python/plugins/potreecraft 2.0
Plugin was not deleted: No such file or directory
Deploying to
/home/troll4hire/.local/share/QGIS/QGIS3/profiles/defau
lt/python/plugins/potreecraft 2.0
Compiling to make sure install is clean
Compiled 0 UI files
Skipping resources.qrc (unchanged)
Compiled 0 resource files
No help directory exists in the current directory
Copying potreecraft ----> ERROR
Copying 2.0.py ----> ERROR
Copying potreecraft ----> ERROR
Copying 2.0_dialog.py ----> ERROR
Copying __init__.py
Copying potreecraft ----> ERROR
Copying 2.0_dialog_base.ui ----> ERROR
Copying resources.py
Copying icon.png
Copying metadata.txt
No help files configured---skipping...

ERRORS:
Error copying files: potreecraft, No such file or directory
Error copying files: 2.0.py, No such file or directory
Error copying files: potreecraft, No such file or directory
Error copying files: 2.0_dialog.py, No such file or
directory
Error copying files: potreecraft, No such file or directory
Error copying files: 2.0_dialog_base.ui, No such file or
directory

One or more files/directories specified in your config file
failed to deploy---make sure they exist or if not needed
remove
```

them from the config. To ensure proper deployment, make sure your
UI and resource files are compiled. Using dclean to delete the
plugin before deploying may also help.

Thought for 20s

The problem is your **internal plugin name contains a space and a version string**.

`pb_tool` reads file lists in `pb_tool.cfg` as **space-separated names**, so it is splitting

ChatGPT Auto

- `potreecraft 2.0_dialog.py` into `potreecraft` and `2.0_dialog.py`

That matches your log exactly:

- `Copying potreecraft ----> ERROR`
- `Copying 2.0.py ----> ERROR`
- `Copying 2.0_dialog.py ----> ERROR`

`pb_tool` expects a config with entries like `python_files: __init__.py geosys.py geosys_dockwidget.py`, i.e. whitespace-delimited filenames, and the plugin `name` is the directory created under the QGIS plugins folder. GitHub +1

So the fix is:

Use a clean internal name

Your **plugin folder/module/class file base name** should be something like:

Plain text

not:

Plain text

For QGIS plugins, the internal package/folder name should be a Python-friendly identifier. In practice, use lowercase letters, numbers, and underscores only. Tutorials

and plugin-builder guidance follow that convention, while the human-readable plugin title is separate. QGIS Tutorials +1

Rename these files

Rename:

```plain text
potreecraft 2.0_dialog_base.ui    -> potreecraft_dialog_base.ui
```

And make sure the plugin directory itself is also renamed from something like:

```plain text
potreecraft 2.0
```

to:

```plain text
potreecraft
```

Fix `pb_tool.cfg`

Open `pb_tool.cfg` and change the internal plugin name and file lists.

It should look roughly like this:

```ini
[plugin]
name: potreecraft
plugin_path:

[files]
python_files: __init__.py potreecraft.py potreecraft_dialog.py
main_dialog: potreecraft_dialog_base.ui
resource_files: resources.py
extras: icon.png metadata.txt
```

Depending on your template, the exact section names may vary a bit, but the important part is: **no spaces in internal filenames**. `pb_tool` requires a config file declaring the plugin files it deploys. G Sherman +1

Keep the display name separate

If you want the plugin to appear in QGIS as:

```plain text
PotreeCraft 2.0
```

put that in `metadata.txt`, not in the Python/package/file names.

For example in `metadata.txt`:

```ini
name=PotreeCraft 2.0
version=2.0
```

That way:

- internal/package name = `potreecraft`
- displayed plugin name = `PotreeCraft 2.0`

Then clean and deploy

After renaming:

```bash
pb_tool dclean
pb_tool deploy --quick
```

If that works, then later:

```bash
pb_tool compile
pb_tool deploy
```

when you start using `.ui` or `.qrc` changes.

Why this happened

Your log already proves `pb_tool` is tokenizing filenames on spaces, because it tried to copy `potreecraft` and `2.0.py` as separate files. That is the core issue, not QGIS itself.

Paste your current `pb_tool.cfg` and I'll rewrite it into a working version.

Sources